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Title: Smart Arrhythmia Detection system using ECG and PPG signals.

Abstract

Cardiac arrhythmias are among the leading causes of cardiovascular complications, making early and continuous monitoring essential for timely diagnosis and intervention. While electrocardiography (ECG) remains the gold standard for cardiac rhythm analysis, photoplethysmography (PPG) offers a convenient and non-invasive means of monitoring blood volume changes. This work proposes a multimodal arrhythmia detection system that combines ECG and PPG signals to improve the accuracy and reliability of cardiac abnormality detection. The proposed framework acquires synchronized ECG and PPG signals, along with morphological and statistical features that characterize normal and abnormal cardiac activity. Pattern recognition techniques can be employed to identify distinctive ECG and PPG waveform characteristics associated with different arrhythmias. These features will be subsequently analyzed to classify heart rhythms as normal or arrhythmic. In addition to arrhythmia detection, the system continuously estimates and displays the user's heart rate in real time, providing an intuitive interface for ongoing cardiac monitoring. Our proposed system aims to enhance diagnostic performance while enabling a user-friendly solution suitable for continuous healthcare monitoring by analyzing ECG and PPG patterns that show arrhythmic patterns. Our proposed system will be able to support early detection of cardiac abnormalities and facilitate remote patient monitoring in making informed diagnostic decisions.

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